

Artificial Intelligence Basics Course Summary

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1 Key Terms

1.1 Search Algorithms

- **Minimax Search:** A search algorithm for two-player zero-sum games that aims to find the best move for the current player, assuming the opponent also plays optimally.
- **Alpha-Beta Pruning:** A technique to reduce the search space in Minimax search by avoiding the evaluation of branches that cannot affect the optimal decision.
- **Game Tree:** A tree-like structure representing all possible game states and moves, with leaf nodes representing outcomes.

1.2 Neural Networks

- **Perceptron:** A fundamental unit in neural networks that takes multiple inputs, applies weights to them, sums them, and produces an output based on a threshold.

1.3 Machine Learning Paradigms

- **Turing Test:** An operational test for intelligent behavior, the Imitation Game, where an interrogator tries to distinguish between a human and a machine based on their responses.
- **Reinforcement Learning:** A type of machine learning where an agent learns to make decisions by interacting with an environment and receiving rewards or penalties.

2 Problem Types

2.1 AI Foundations

- **Understanding the Nature of Intelligence:** This problem explores the definition and characteristics of intelligence, both human and artificial. Different perspectives and definitions of AI are examined.
- **Overcoming the Limits of Search:** This problem addresses the computational challenges of exhaustive search in complex games like chess and Go. The exponential growth of search space is highlighted, leading to the need for machine learning techniques to supplement search.

2.2 AI System Architecture

- **Understanding the Components of AI Systems:** This problem focuses on identifying the key components of AI algorithms, including knowledge representation, decision-making processes, and the ability to generalize to new situations.

2.3 AI Interoperability

- **Bridging the Gap Between Different AI Domains:** This problem investigates the commonalities and differences between seemingly disparate AI tasks, such as game playing and speech synthesis. The shared algorithmic components and challenges are analyzed.

2.4 AI Performance Measurement

- **Evaluating Intelligent Behavior:** This problem focuses on developing methods for assessing whether a machine exhibits intelligent behavior. The Turing Test and its limitations are discussed, along with counterarguments like Searle's Chinese Room argument.

2.5 AI Challenges and Risks

- **Addressing AI Safety and Ethical Concerns:** This problem explores the potential risks and ethical implications of AI systems, including bias in algorithms and the vulnerability of neural networks to adversarial attacks.
- **Defining the Challenges of Autonomous Systems:** This problem explores the complexities of building autonomous systems, using self-driving cars as a prime example. The reasons for choosing autonomous vehicles as an AI Grand Challenge are examined.

3 Algorithm Solutions

3.1 Game Playing AI

- **Minimax Search:** A recursive algorithm for two-player zero-sum games that explores the game tree to find the best move for the current player, assuming the opponent also plays optimally. It alternates between maximizing the score for the current player and minimizing the score for the opponent.
- **Alpha-Beta Pruning:** An optimization technique for Minimax search that reduces the search space by avoiding the evaluation of branches that cannot affect the optimal decision. It uses alpha and beta values to prune branches.